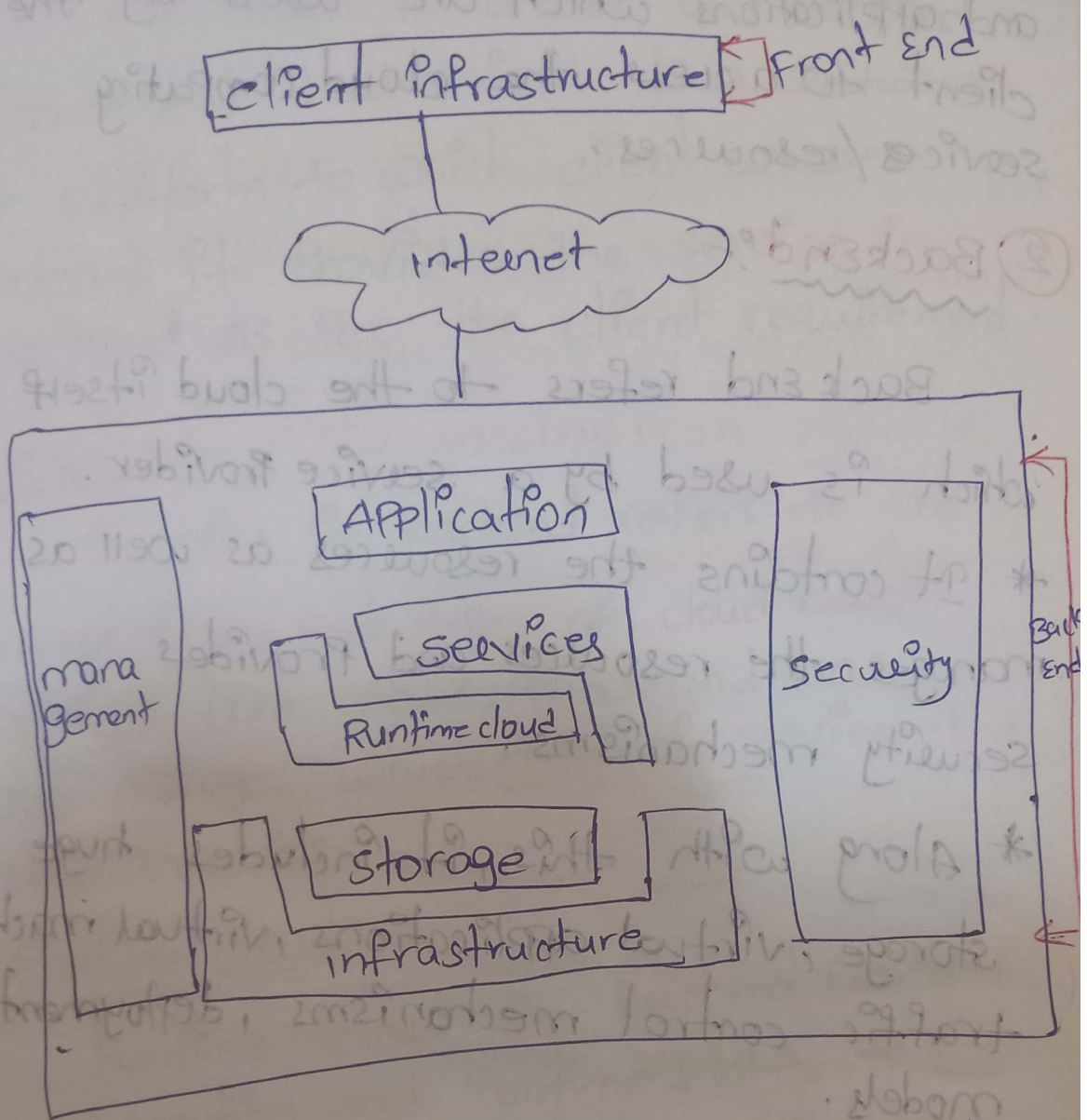


cloud computing architecture :-



The cloud architecture is divided into 2 Parts.

- ① Front end
- ② Back end

① Front end :-

Front end of the cloud architecture refers to the client side of cloud computing.

* Means it contains all user interfaces and applications which are used by the client to access the cloud computing services/resources.

② Backend:-

Backend refers to the cloud itself which is used by a service provider.

* It contains the resources as well as manages the resources and provides security mechanisms.

* Along with this, it includes huge storage, virtual applications, virtual machines, traffic control mechanisms, deployment models.

Components of cloud computing architecture

① client infrastructure:-

client infrastructure is a part of the front end component. It contains the applications and user interfaces which are required to access the cloud platforms. In other words, it provides a GUI to interact with the cloud.

② Applications:-

Application is a part of backend component that refers to a software or platform to which client accesses. means it provides the service in backend as per the client requirement.

③ Service:-

Service in backend refers to the major three part types of cloud based services like SaaS, PaaS and IaaS. also manages which type of service the user accesses.

④ Runtime cloud:-

Runtime cloud in backend provides the execution and Runtime Platform/ environment to the virtual machine.

⑤ Storage:-

Storage in backend provides flexible and scalable storage service and management of stored data.

⑥ Infrastructure:-

cloud infrastructure in backend refers to management of backend components like application, service, runtime, storage, infrastructure, and other security mechanisms etc--

⑥ Infrastructure:-

~~Not Infrastructure~~

cloud infrastructure in backend refers to the hardware and software components of cloud like it includes servers, storage, network devices, virtualized software etc.

⑦ Management:-

management in backend refers to management of backend components like application, service, runtime cloud, storage, infrastructure, and other security mechanisms etc--

⑧ Security:-

security in backend refers to implementation of different security mechanisms in the backend for secure cloud resources, systems, files, and infrastructure to end-users.

⑨ Internet:-

internet connection acts as a medium or a bridge between frontend and backend and establishes the interaction and communication between frontend and backend.

⑩ Database:-

Database in backend refers to provide database for storing structured data, such as SQL and NoSQL database.

⑪ Networking:-

Networking in backend services that provide networking infrastructure for application in the cloud, such as load balancing, DNS and virtual private network.

(12) Analytics:-

Analytics backend service that provides analytics capabilities for data in the cloud such as warehouse, business intelligence and machine learning.

Layers:-

(1) Physical layers:-

* The foundational layer, consisting of the physical hardware such as servers, storage, and networking equipment.

(2) Virtualization layer:-

sits on top of the hardware, using software to create virtual versions of resources for flexible and efficient storage.